



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**FINAL EXAMINATION
SEMESTER III, SESSION 2020/2021
PART I**

COURSE CODE	:	FSPC 0034
COURSE NAME	:	CHEMISTRY II
PROGRAMME	:	ASASI UTMSPACE
DURATION	:	1 HOUR 30 MINUTES
DATE	:	15 APRIL 2021

INSTRUCTION TO CANDIDATES:

1. DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DOSO.
2. THERE ARE **FOUR (4)** QUESTIONS. ANSWER ALL QUESTIONS.

WARNING!

Students caught copying/cheating during the examination will be liable for disciplinary actions and UTMSPACE may recommend the student to be expelled from the study.

This booklet consists of (8) printed pages including this page.

QUESTION 1 (10 MARKS)

- a) Magnesium reacted with oxygen to form the compound magnesium chloride. The enthalpy changes associated with the reaction are given as below (in kJ mol^{-1}).

Enthalpy of atomization of magnesium = +146

First ionization energy of magnesium = +736

Second ionization energy of magnesium = + 1450

Enthalpy of atomization of chloride = +121

First electron affinity of chloride = -364

Enthalpy of formation of magnesium chloride = -658

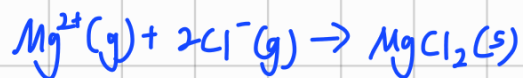
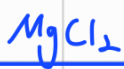
- i. Based on the information given, construct a Born-Haber cycle. Please include all the energies and chemical equations involved.
- ii. Write the Born-Haber cycle equation for magnesium chloride.
Calculate the lattice energy of magnesium chloride.

(5 + 2 marks)

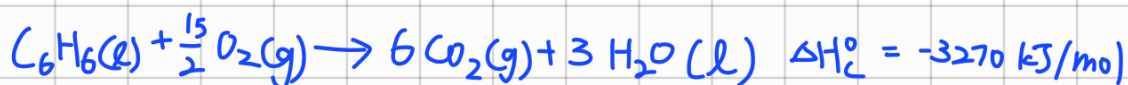
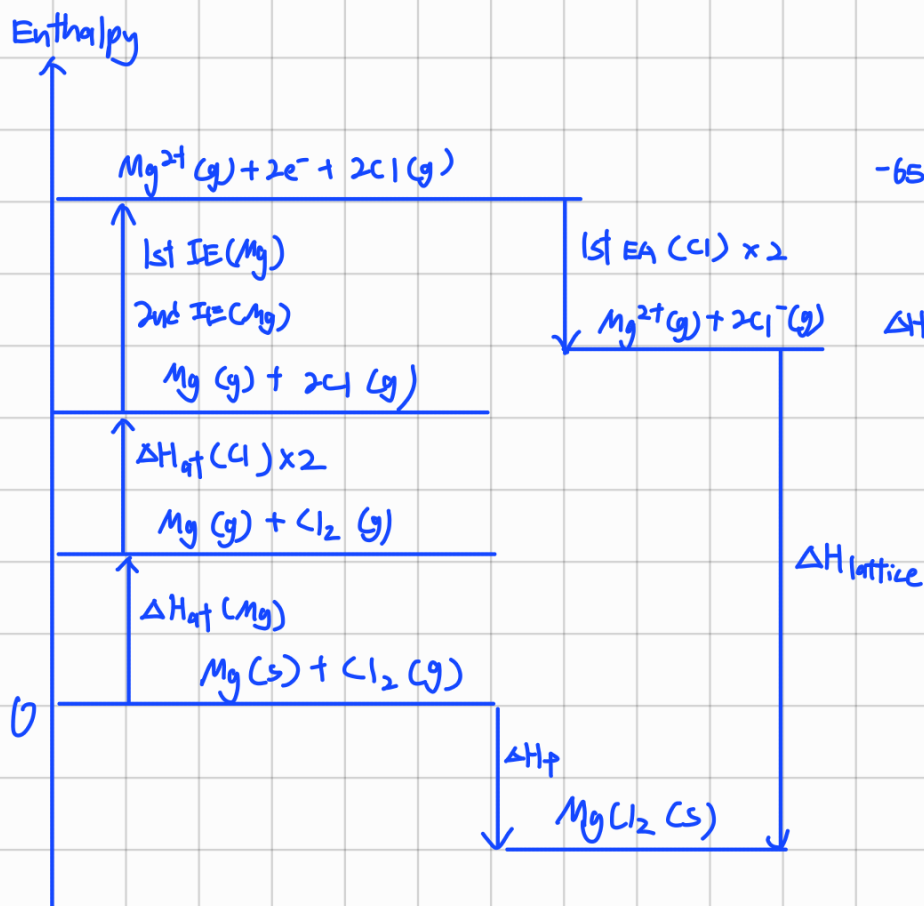
- b) The standard enthalpies for benzene, hydrogen and carbon are $-3270 \text{ kJ mol}^{-1}$, -286 kJ mol^{-1} and -394 kJ mol^{-1} respectively.

- i. Write the thermochemical equation for the combustion of liquid benzene.
- ii. Write the standard enthalpy equation of this reaction.
Determine the standard enthalpy of formation of benzene.

(1 + 2 marks)

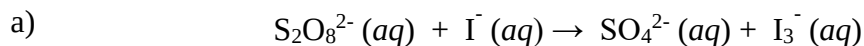


$$\begin{aligned}\Delta H_f &= \Delta H_{\text{at}}(\text{Mg}) + \Delta H_{\text{at}}(\text{Cl}) + \\ &\quad \Delta H_{1\text{st IE}}(\text{Mg}) + \Delta H_{2\text{nd IE}}(\text{Mg}) + \\ &\quad \Delta H_{1\text{st EA}}(\text{Cl}) + \Delta H_{\text{lattice}} \\ -658 &= +146 + 2(+121) + 736 + 1450 + \\ &\quad 2(-364) + \Delta H_{\text{lattice}} \\ \Delta H_{\text{lattice}} &= -2504 \text{ kJ/mol}\end{aligned}$$



$$\Delta H = -3270 - (-286 + (-394))$$

$$= -2590 \text{ kJ/mol}$$

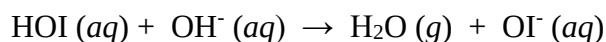
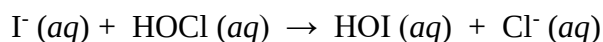
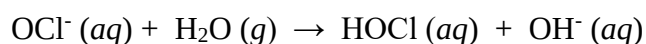
QUESTION 2 (10 MARKS)

The equation above describes the reaction between peroxydisulfate ion, ($\text{S}_2\text{O}_8^{2-}$) and iodide ion, (I^-) at 330 K, in which peroxydisulfate ion is first order while iodide ion is second order kinetics. When $[\text{S}_2\text{O}_8^{2-}]$ is 0.256 M and $[\text{I}^-]$ is 0.583 M, the rate of reaction is determined as $2.6 \times 10^{-5} \text{ Ms}^{-1}$.

- i. Calculate the rate constant value of the reaction.
- ii. Determine the rate constant value when $[\text{S}_2\text{O}_8^{2-}] = 0.050 \text{ M}$ and $[\text{I}^-] = 0.113 \text{ M}$.

(2 + 1 marks)

- b) Consider the multi-step equations of a catalytic reaction below.



- i. Based on the chemical equations, list all the intermediates in the catalytic reaction.
- ii. Write the overall equation for the reaction.
- iii. Given that the first step is the rate limiting step, identify the rate law for the overall reaction.

(1 + 1 + 1marks)

- c) Consider a reaction between ethyl acetate, $\text{CH}_3\text{COOC}_2\text{H}_5$ and hydroxide ion, OH^- . It was found that the order of reaction is first order with respect to $\text{CH}_3\text{COOC}_2\text{H}_5$ and second order with respect to OH^- . Given that the activation energy for the reaction is $3.4 \times 10^2 \text{ kJmol}^{-1}$ and at 25°C , the rate of reaction is $6.5 \times 10^{-4} \text{ Ms}^{-1}$ when $[\text{CH}_3\text{COOC}_2\text{H}_5] = 1.5 \text{ M}$ and $[\text{OH}^-] = 2.1 \text{ M}$. Determine the rate of reaction at a temperature of 50°C .

(4 marks)

$$2) (a) (i) R = k[S_{2O_8}^{2-}]^1[I^-]^2$$

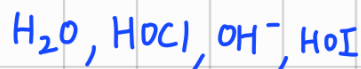
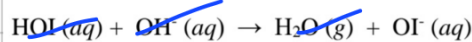
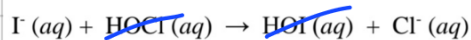
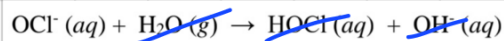
$$2.6 \times 10^{-5} = k(0.256)(0.583)^2$$

$$k = 2.988 \times 10^{-4} s^{-1}$$

$$(ii) 2.6 \times 10^{-5} = k(0.050)(0.113)^2$$

$$k = 0.041 s^{-1}$$

(b) (i) Intermediate:



$$(iii) \text{Rate law: } k[OCl^-][H_2O]$$

order of reaction is second order

$$(c) \ln\left(\frac{k_2}{k_1}\right) = \frac{E_a}{R} \left(\frac{T_2 - T_1}{T_2 \times T_1}\right)$$

$$R = k[CH_3COOC_2H_5]^1[OH^-]^2$$

$$6.5 \times 10^{-4} = k(1.5)(2.1)^2$$

$$k = 9.826 \times 10^{-5} s^{-1}$$

$$\ln \frac{k_2}{9.826 \times 10^{-5}} = \frac{3.4 \times 10^2}{8.314} \left(\frac{323 - 298}{323 \times 298} \right)$$

$$k_2 = e^{0.0106} \times 9.826 \times 10^{-5}$$

$$k_2 = 9.931 \times 10^{-5} s^{-1}$$

$$R = 9.931 \times 10^{-5} (1.5)(2.1)^2$$

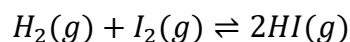
$$= 6.57 \times 10^{-4} M s^{-1}$$

$$\ln x = y$$

$$x = e^y$$

QUESTION 3 (10 MARKS)

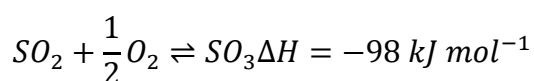
- a) Hydrogen reacts with iodine to form hydrogen iodide.



When 43.0 g of iodine and 0.8 g of hydrogen were heated until equilibrium was established at 470°C, the equilibrium mixture was found to contain 1.7 g of iodine. Determine the equilibrium constant for this reaction.

(4 marks)

- b) The critical step in the manufacture of sulphuric acid is the oxidation of sulphur dioxide.



Explain why the reaction is carried out at 425°C to make the process as economic as possible.

(3 marks)

- c) Often recognized as fluorite or fluorspar, calcium fluoride (CaF₂) is the major source of fluorine in the manufacture of hydrogen fluoride and calcium sulfate. Identify the solubility (in g L⁻¹) of calcium fluoride (CaF₂) in a solution containing 0.005 mol L⁻¹ sodium fluoride solution.

(K_{sp} of CaF₂ = $3.4 \times 10^{-11} \text{ mol}^3 \text{ L}^{-9}$)

soluble salt solution
to give [F⁻] initial

(3marks)

3) (a) $m_{I_2} = 43g$, $m_{H_2} = 0.8g$, $T = 470^\circ C$, $m_{HI} = 1.7g$
 $\text{mol of } I_2 = \frac{43}{254} = 0.17 \text{ mol}$ $\text{mol of } H_2 = \frac{0.8}{2} = 0.4 \text{ mol}$ $\text{mol of } HI = \frac{1.7}{128} = 0.0133$
 $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

0.17	0.4	0
-0.0133	-0.0133	+0.0133
0.1567	0.3867	0.0133

$$K_c = \frac{[HI]^2}{[I_2][H_2]} = \frac{0.0133^2}{(0.1567)(0.3867)} = 0.002919$$

(c) $CaF_2(s) \rightleftharpoons Ca^{2+}(aq) + 2F^{-}$ $K_{sp} = [Ca^{2+}][F^{-}]^2 = 3.4 \times 10^{-11}$

I	-	0	0.005	$x(0.005 + 2x) = 3.4 \times 10^{-11}$
C	-	+x	+2x	$x(0.005) = 3.4 \times 10^{-11}$
E	-	x	0.005 + 2x	$x = 6.8 \times 10^{-9} M$

solubility $= 6.8 \times 10^{-9} (40 + 19)$
 $= 4.012 \times 10^{-7} g L^{-1}$ &

QUESTION 4 (10 MARKS)

- a) H_2PO_4^- is an acid and will react with water.
- Write the chemical equation for the reaction of H_2PO_4^- with water.
 - Using H_2PO_4^- as the weak acid, identify the other components to make a buffer solution.

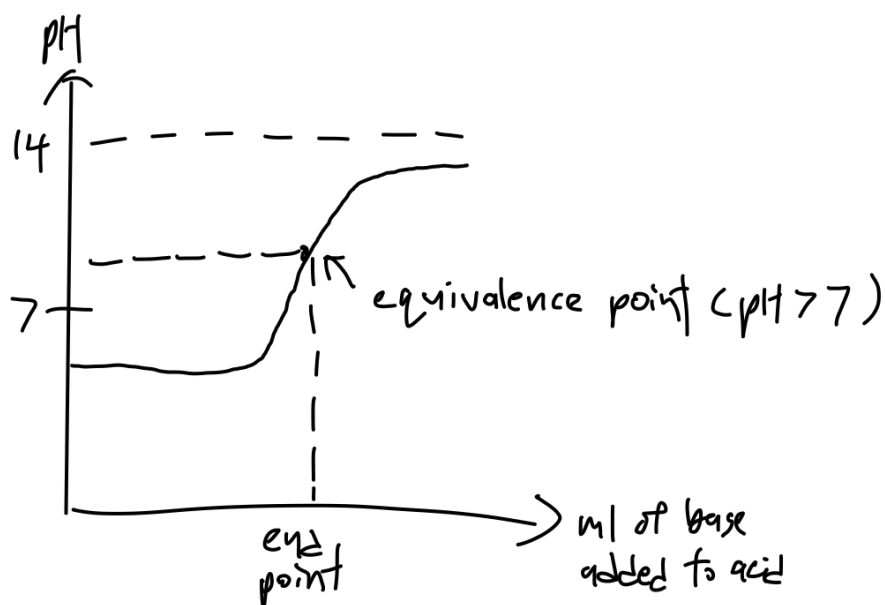
(1 + 1 marks)

- b) i. Methylamine has a pH of 9.5. Determine the concentration of H^+ .
- If hypochlorous acid, HClO is reacted with methylamine, predict the pH at equivalence point.
 K_a of $\text{HClO} = 3.5 \times 10^{-8}$
 K_b of methylamine = 4.4×10^{-4}

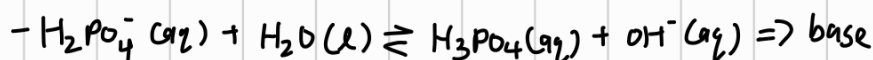
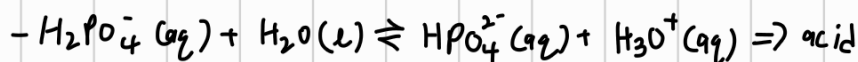
(1 + 1 marks)

- c) Boric acid, H_3BO_3 is commonly used as an antiseptic. A 20 mL 0.10 M solution of boric acid is titrated with 0.20 M sodium hydroxide, NaOH solution (titrant).
 (Given $K_a = 5.7 \times 10^{-10}$, $K_w = 1 \times 10^{-14}$)

- Plot the expected titration curve of the acid base titration.
- Calculate pH of the solution after the addition of 30.00 mL 0.20 M NaOH solution.



4) (a) (i) Since H_2PO_4^- is amphiprotic:



(ii) buffer $\rightarrow$ weak acid + salt of conjugate base

conjugate base: HPO_4^{2-}

salt: Na_2HPO_4

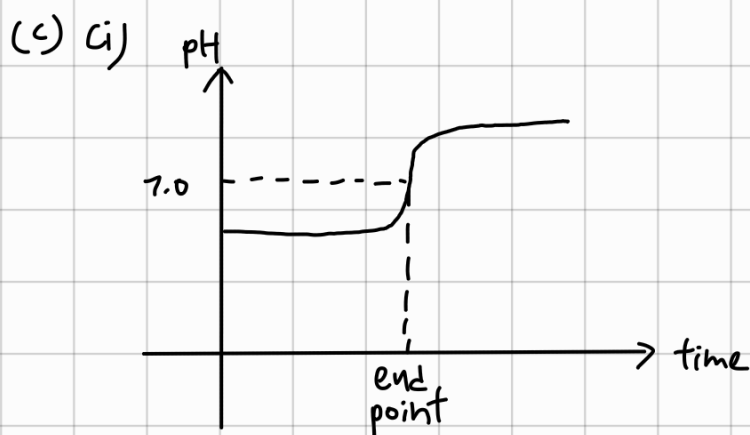
(b) (i) $\text{pH} = -\log[\text{H}^+]$

$$9.5 = -\log[\text{H}^+]$$

$$[\text{H}^+] = 10^{-9.5}$$

$$= 3.16 \times 10^{-10}$$

(ii) Since HClO is weak acid and methylamine, CH_3NH_2 is weak base the pH depends on K_a and K_b . In this situation, $K_b > K_a$ so the pH of the solution > 7 .



(ii) mol of $\text{H}_3\text{BO}_3 = 0.02 \times 0.1 = 2 \times 10^{-3} \text{ mol}$

Volume of NaOH needed at equivalence = $\frac{0.02(0.1)}{0.2} = 0.01 \text{ L}$



B	2×10^{-3}	6×10^{-3}	0
A	-2×10^{-3}	-2×10^{-3}	$+2 \times 10^{-3}$
A	0	4×10^{-3}	2×10^{-3}

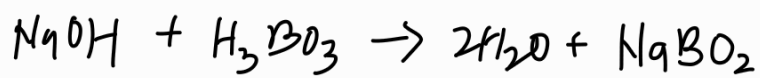
$$K_b = \frac{1 \times 10^{-14}}{5.7 \times 10^{-10}} = 1.754 \times 10^{-5}$$

$$1.754 \times 10^{-5} = \frac{[\text{OH}^-]^2 [\text{H}_3\text{BO}_3]^4}{[\text{B}_4\text{O}_7^{2-}]} = \frac{(2x)^2 (4x)^4}{0.333 - x} = \frac{4x^2 (256x^4)}{0.333}$$

$$x = 0.0422$$

$$\text{pOH} = -\log[\text{OH}^-] = -\log[0.0422] = 1.37, \text{ pH} = 12.63$$

(ii)



B	6×10^{-3}	2×10^{-3}	0
A	-2×10^{-3}	-2×10^{-3}	$+2 \times 10^{-3}$
A	4×10^{-3}		2×10^{-3}

$$[\text{NaOH}] \equiv [\text{OH}^-] = \frac{4 \times 10^{-3}}{0.05} = 0.08$$

$$\text{pOH} = -\log(0.08)$$

$$= 1.10$$

$$\text{pH} = 12.9$$

$$\frac{M_a V_a}{M_b V_b} = \frac{a}{b}$$

$V_{\text{base}} = 10 \text{ ml}$ at equivalence point

$$\begin{aligned} \text{mol of NaOH added} &= 0.03(0.2) \\ &= 0.006 \text{ mol} \end{aligned}$$

$$\begin{aligned} \text{mol of H}_3\text{BO}_3 &= 0.02(0.1) \\ &= 0.002 \text{ mol} \end{aligned}$$

Appendix1 : THE PERIODIC TABLE

1	2	13	14	15	16	17	18
IA	IIA	IIIA	IVA	VA	VIA	VIIA	VIIIA
1 H 1.01	2 He 4.00	5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
3 Li 6.94	4 Be 9.01	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
11 Na 22.99	12 Mg 24.31	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
19 K 39.1	20 Ca 40.08	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.6	53 I 126.9	54 Xe 131.29
37 Rb 85.47	38 Sr 87.62	81 Tl 204.4	82 Pb 207.2	83 Bi 209	84 Po (209)	85 At (210)	86 Rn (222)
55 Cs 132.9	56 Ba 137.3	89 Ac[^] (227)	90 Th 232.0	91 Pa (231)	92 U 238.0	93 Np (237)	94 Pu (244)
87 Fr (223)	88 Ra (226)	101 La[*] 138.9	102 Ce 140.1	103 Pr 140.9	104 Nd 144.2	105 Pm (145)	106 Sm 150.4
		107 Eu 152.0	108 Gd 157.3	109 Tb 158.9	110 Dy 162.5	111 Ho 164.9	112 Er 167.3
		113 Yb 173.0	114 Lu 175.0	115 Yt 176.9	116 La 178.9	117 Ce 180.9	118 Pr 182.9
		119 Sm 183.9	120 Eu 186.2	121 Gd 188.9	122 Tb 190.9	123 Dy 193.2	124 Ho 195.1
		125 Er 197.0	126 Tm 198.9	127 Yb 200.0	128 Lu 200.9	129 La 201.1	130 Ce 201.2
		131 Pr 201.3	132 Nd 201.6	133 Pm (202)	134 Sm 202.6	135 Eu 203.7	136 Gd 204.7
		137 Tb 204.4	138 Dy 206.4	139 Ho 207.2	140 Er 208.1	141 Tm 209.1	142 Yb 209.1
		143 Lu 209.1	144 La 209.1	145 Ce 209.1	146 Pr 209.1	147 Nd 209.1	148 Pm (209)
		149 Sm 207.2	150 Eu 207.2	151 Gd 207.2	152 Tb 207.2	153 Dy 207.2	154 Ho 207.2
		155 Er 207.2	156 Tm 207.2	157 Yb 207.2	158 Lu 207.2	159 La 207.2	160 Ce 207.2
		161 Pr 207.2	162 Nd 207.2	163 Pm (207)	164 Sm 207.2	165 Eu 207.2	166 Gd 207.2
		167 Tb 207.2	168 Dy 207.2	169 Ho 207.2	170 Er 207.2	171 Tm 207.2	172 Yb 207.2
		173 Lu 207.2	174 La 207.2	175 Ce 207.2	176 Pr 207.2	177 Nd 207.2	178 Pm (207)
		179 Sm 207.2	180 Eu 207.2	181 Gd 207.2	182 Tb 207.2	183 Dy 207.2	184 Ho 207.2
		185 Er 207.2	186 Tm 207.2	187 Yb 207.2	188 Lu 207.2	189 La 207.2	190 Ce 207.2
		191 Pr 207.2	192 Nd 207.2	193 Pm (207)	194 Sm 207.2	195 Eu 207.2	196 Gd 207.2
		197 Tb 207.2	198 Dy 207.2	199 Ho 207.2	200 Er 207.2	201 Tm 207.2	202 Yb 207.2
		203 Lu 207.2	204 La 207.2	205 Ce 207.2	206 Pr 207.2	207 Nd 207.2	208 Pm (207)
		209 Sm 207.2	210 Eu 207.2	211 Gd 207.2	212 Tb 207.2	213 Dy 207.2	214 Ho 207.2
		215 Er 207.2	216 Tm 207.2	217 Yb 207.2	218 Lu 207.2	219 La 207.2	220 Ce 207.2
		221 Pr 207.2	222 Nd 207.2	223 Pm (207)	224 Sm 207.2	225 Eu 207.2	226 Gd 207.2
		227 Tb 207.2	228 Dy 207.2	229 Ho 207.2	230 Er 207.2	231 Tm 207.2	232 Yb 207.2
		233 Lu 207.2	234 La 207.2	235 Ce 207.2	236 Pr 207.2	237 Nd 207.2	238 Pm (207)
		239 Sm 207.2	240 Eu 207.2	241 Gd 207.2	242 Tb 207.2	243 Dy 207.2	244 Ho 207.2
		245 Er 207.2	246 Tm 207.2	247 Yb 207.2	248 Lu 207.2	249 La 207.2	250 Ce 207.2
		251 Pr 207.2	252 Nd 207.2	253 Pm (207)	254 Sm 207.2	255 Eu 207.2	256 Gd 207.2
		257 Tb 207.2	258 Dy 207.2	259 Ho 207.2	260 Er 207.2	261 Tm 207.2	262 Yb 207.2
		263 Lu 207.2	264 La 207.2	265 Ce 207.2	266 Pr 207.2	267 Nd 207.2	268 Pm (207)
		269 Sm 207.2	270 Eu 207.2	271 Gd 207.2	272 Tb 207.2	273 Dy 207.2	274 Ho 207.2
		275 Er 207.2	276 Tm 207.2	277 Yb 207.2	278 Lu 207.2	279 La 207.2	280 Ce 207.2
		281 Pr 207.2	282 Nd 207.2	283 Pm (207)	284 Sm 207.2	285 Eu 207.2	286 Gd 207.2
		287 Tb 207.2	288 Dy 207.2	289 Ho 207.2	290 Er 207.2	291 Tm 207.2	292 Yb 207.2
		293 Lu 207.2	294 La 207.2	295 Ce 207.2	296 Pr 207.2	297 Nd 207.2	298 Pm (207)
		299 Sm 207.2	300 Eu 207.2	301 Gd 207.2	302 Tb 207.2	303 Dy 207.2	304 Ho 207.2
		305 Er 207.2	306 Tm 207.2	307 Yb 207.2	308 Lu 207.2	309 La 207.2	310 Ce 207.2
		311 Pr 207.2	312 Nd 207.2	313 Pm (207)	314 Sm 207.2	315 Eu 207.2	316 Gd 207.2
		317 Tb 207.2	318 Dy 207.2	319 Ho 207.2	320 Er 207.2	321 Tm 207.2	322 Yb 207.2
		323 Lu 207.2	324 La 207.2	325 Ce 207.2	326 Pr 207.2	327 Nd 207.2	328 Pm (207)
		329 Sm 207.2	330 Eu 207.2	331 Gd 207.2	332 Tb 207.2	333 Dy 207.2	334 Ho 207.2
		335 Er 207.2	336 Tm 207.2	337 Yb 207.2	338 Lu 207.2	339 La 207.2	340 Ce 207.2
		341 Pr 207.2	342 Nd 207.2	343 Pm (207)	344 Sm 207.2	345 Eu 207.2	346 Gd 207.2
		347 Tb 207.2	348 Dy 207.2	349 Ho 207.2	350 Er 207.2	351 Tm 207.2	352 Yb 207.2
		353 Lu 207.2	354 La 207.2	355 Ce 207.2	356 Pr 207.2	357 Nd 207.2	358 Pm (207)
		359 Sm 207.2	360 Eu 207.2	361 Gd 207.2	362 Tb 207.2	363 Dy 207.2	364 Ho 207.2
		365 Er 207.2	366 Tm 207.2	367 Yb 207.2	368 Lu 207.2	369 La 207.2	370 Ce 207.2
		371 Pr 207.2	372 Nd 207.2	373 Pm (207)	374 Sm 207.2	375 Eu 207.2	376 Gd 207.2
		377 Tb 207.2	378 Dy 207.2	379 Ho 207.2	380 Er 207.2	381 Tm 207.2	382 Yb 207.2
		383 Lu 207.2	384 La 207.2	385 Ce 207.2	386 Pr 207.2	387 Nd 207.2	388 Pm (207)
		389 Sm 207.2	390 Eu 207.2	391 Gd 207.2	392 Tb 207.2	393 Dy 207.2	394 Ho 207.2
		395 Er 207.2	396 Tm 207.2	397 Yb 207.2	398 Lu 207.2	399 La 207.2	400 Ce 207.2
		401 Pr 207.2	402 Nd 207.2	403 Pm (207)	404 Sm 207.2	405 Eu 207.2	406 Gd 207.2
		407 Tb 207.2	408 Dy 207.2	409 Ho 207.2	410 Er 207.2	411 Tm 207.2	412 Yb 207.2
		413 Lu 207.2	414 La 207.2	415 Ce 207.2	416 Pr 207.2	417 Nd 207.2	418 Pm (207)
		419 Sm 207.2	420 Eu 207.2	421 Gd 207.2	422 Tb 207.2	423 Dy 207.2	424 Ho 207.2
		425 Er 207.2	426 Tm 207.2	427 Yb 207.2	428 Lu 207.2	429 La 207.2	430 Ce 207.2
		431 Pr 207.2	432 Nd 207.2	433 Pm (207)	434 Sm 207.2	435 Eu 207.2	436 Gd 207.2
		437 Tb 207.2	438 Dy 207.2	439 Ho 207.2	440 Er 207.2	441 Tm 207.2	442 Yb 207.2
		443 Lu 207.2	444 La 207.2	445 Ce 207.2	446 Pr 207.2	447 Nd 207.2	448 Pm (207)
		449 Sm 207.2	450 Eu 207.2	451 Gd 207.2	452 Tb 207.2	453 Dy 207.2	454 Ho 207.2
		455 Er 207.2	456 Tm 207.2	457 Yb 207.2	458 Lu 207.2	459 La 207.2	460 Ce 207.2
		461 Pr 207.2	462 Nd 207.2	463 Pm (207)	464 Sm 207.2	465 Eu 207.2	466 Gd 207.2
		467 Tb 207.2	468 Dy 207.2	469 Ho 207.2	470 Er 207.2	471 Tm 207.2	472 Yb 207.2
		473 Lu 207.2	474 La 207.2	475 Ce 207.2	476 Pr 207.2	477 Nd 207.2	478 Pm (207)
		479 Sm 207.2	480 Eu 207.2	481 Gd 207.2	482 Tb 207.2	483 Dy 207.2	484 Ho 207.2
		485 Er 207.2	486 Tm 207.2	487 Yb 207.2	488 Lu 207.2	489 La 207.2	490 Ce 207.2
		491 Pr 207.2	492 Nd 207.2	493 Pm (207)	494 Sm 207.2	495 Eu 207.2	496 Gd 207.2
		497 Tb 207.2	498 Dy 207.2	499 Ho 207.2	500 Er 207.2	501 Tm 207.2	502 Yb 207.2
		503 Lu 207.2	504 La 207.2	505 Ce 207.2	506 Pr 207.2	507 Nd 207.2	508 Pm (207)
		509 Sm 207.2	510 Eu 207.2	511 Gd 207.2	512 Tb 207.2	513 Dy 207.2	514 Ho 207.2
		515 Er 207.2	516 Tm 207.2	517 Yb 207.2	518 Lu 207.2	519 La 207.2	520 Ce 207.2
		521 Pr 207.2	522 Nd 207.2	523 Pm (207)	524 Sm 207.2	525 Eu 207.2	526 Gd 207.2
		527 Tb 207.2	528 Dy 207.2	529 Ho 207.2	530 Er 207.2	531 Tm 207.2	532 Yb 207.2
		533 Lu 207.2	534 La 207.2	535 Ce 207.2	536 Pr 207.2	53	

Appendix 2 :LIST OF SELECTED CONSTANT VALUES

Speed of light in a vacuum	c	=	$3.0 \times 10^8 \text{ m s}^{-1}$
Avogadro's number	N_A	=	$6.02 \times 10^{23} \text{ mol}^{-1}$
Faraday constant	F	=	$9.65 \times 10^4 \text{ C mol}^{-1}$
Planck constant	h	=	$6.6256 \times 10^{-34} \text{ J s}$
Reduced Planck constant	$\hbar$	=	$1.054 \times 10^{-34} \text{ J s}$
Rydberg constant	R_H	=	$1.097 \times 10^7 \text{ m}^{-1}$ = $2.18 \times 10^{-14} \text{ J}$
Molar of gases constant	R	=	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ = $0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$
Boltzmann constant	k	=	$1.3807 \times 10^{-27} \text{ J K}^{-1}$
Mass of proton	M_p	=	$1.672 \times 10^{-27} \text{ kg}$
Electronic Bohr magneton	m_B	=	$9.2741 \times 10^{-24} \text{ J T}^{-1}$
Nuclear Bohr magneton	b_N	=	$5.05 \times 10^{-27} \text{ J T}^{-1}$
Vapour pressure of water	P_{water}	=	23.8 torr
Electron charge	e^-	=	$1.602 \times 10^{-19} \text{ C}$

UNIT AND CONVERSION FACTOR

Energy	$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 1 \text{ N m} = 10^7 \text{ erg}$ $1 \text{ calorie} = 4.184 \text{ Joule}$ $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$
Pressure	$1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.325 \text{ kPa} = 101325 \text{ N m}^{-2}$

Appendix 3 : STANDARD REDUCTION POTENTIAL (E°_{Red})

$\text{PbO}_2(\text{s}) + \text{SO}_4^{2-}(\text{aq}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{PbSO}_4(\text{s}) + 2\text{H}_2\text{O}(\ell)$	+1.69
$\text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5\text{e}^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\ell)$	+1.51
$\text{Au}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Au}(\text{s})$	+1.50
$\text{ClO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 8\text{e}^- \rightarrow \text{Cl}^-(\text{aq}) + 4\text{H}_2\text{O}(\ell)$	+1.39
$\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-(\text{aq})$	+1.36
$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+(\text{aq}) + 6\text{e}^- \rightarrow 2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}(\ell)$	+1.33
$2\text{HNO}_3(\text{aq}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow \text{N}_2\text{O}(\text{g}) + 3\text{H}_2\text{O}(\ell)$	+1.30
$\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}(\ell)$	+1.23
$\text{MnO}_2(\text{s}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\ell)$	+1.22
$\text{Br}_2(\text{aq}) + 2\text{e}^- \rightarrow 2\text{Br}^-(\text{aq})$	+1.07
$\text{Hg}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Hg}(\ell)$	+0.85
$\text{ClO}^-(\text{aq}) + \text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow \text{Cl}^-(\text{aq}) + 2\text{OH}^-(\text{aq})$	+0.84
$\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag}(\text{s})$	+0.80
$\text{NO}_3^-(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \text{NO}_2(\text{g}) + \text{H}_2\text{O}(\ell)$	+0.80
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{aq})$	+0.77
$\text{O}_2(\text{g}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2\text{O}_2(\ell)$	+0.70
$\text{I}_2(\text{s}) + 2\text{e}^- \rightarrow 2\text{I}^-(\text{aq})$	+0.54
$\text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\ell) + 4\text{e}^- \rightarrow 4\text{OH}^-(\text{aq})$	+0.40
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$	+0.34
$\text{SO}_4^{2-}(\text{aq}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2\text{SO}_3(\text{aq}) + \text{H}_2\text{O}(\ell)$	+0.17
$\text{Sn}^{4+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Sn}^{2+}(\text{aq})$	+0.15
$\text{S}(\text{s}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2\text{S}(\text{aq})$	+0.14
$\text{AgBr}(\text{s}) + \text{e}^- \rightarrow \text{Ag}(\text{s}) + \text{Br}^-(\text{aq})$	+0.07
$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_{2(\text{g})}$	0.00
$\text{Pb}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Pb}(\text{s})$	-0.13
$\text{Sn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Sn}(\text{s})$	-0.14
$\text{AgI}(\text{s}) + \text{e}^- \rightarrow \text{Ag}(\text{s}) + \text{I}^-(\text{aq})$	-0.15
$\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ni}(\text{s})$	-0.26
$\text{Co}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Co}(\text{s})$	-0.28
$\text{PbSO}_4(\text{s}) + 2\text{e}^- \rightarrow \text{Pb}(\text{s}) + \text{SO}_4^{2-}(\text{aq})$	-0.36
$\text{Se}(\text{s}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2\text{Se}(\text{aq})$	-0.40
$\text{Cd}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cd}(\text{s})$	-0.40
$\text{Cr}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Cr}^{2+}(\text{aq})$	-0.41
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Fe}(\text{s})$	-0.45
$\text{NO}_2^-(\text{aq}) + \text{H}_2\text{O}(\ell) + \text{e}^- \rightarrow \text{NO}(\text{g}) + 2\text{OH}^-(\text{aq})$	-0.46
$\text{Ag}_2\text{S}(\text{s}) + 2\text{e}^- \rightarrow 2\text{Ag}(\text{s}) + \text{S}^{2-}(\text{aq})$	-0.69
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$	-0.76
$2\text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$	-0.83
$\text{Cr}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cr}(\text{s})$	-0.91
$\text{Se}(\text{s}) + 2\text{e}^- \rightarrow \text{Se}^{2-}(\text{aq})$	-0.92
$\text{SO}_4^{2-}(\text{aq}) + \text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow \text{SO}_3^{2-}(\text{aq}) + 2\text{OH}^-(\text{aq})$	-0.93
$\text{Al}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Al}(\text{s})$	-1.66
$\text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Mg}(\text{s})$	-2.37
$\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}(\text{s})$	-2.71
$\text{Ca}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ca}(\text{s})$	-2.87
$\text{Ba}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ba}(\text{s})$	-2.91
$\text{Li}^+(\text{aq}) + \text{e}^- \rightarrow \text{Li}(\text{s})$	-3.04